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https://doi.org/10.1007/s12633-025-03360-1

RESEARCH

Polyhedral Oligomeric Silsesquioxane/Multiwalled Carbon Nanotube (MWCNT/POSS) Hybrid Filler in Ethylene-Propylene-Diene Monomer/Styrene-Butadiene Rubber (EPDM/SBR) Nanocomposites: Preparation and Properties

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Received: 4 March 2025 / Accepted: 4 June 2025

Abstract

In this study, ethylene-propylene-diene monomer/styrene-butadiene rubber (EPDM/SBR) nanocomposites were prepared using a two-roll mill mixing method, incorporating a hybrid filler of polyhedral oligomeric silsesquioxane (POSS) and multiwalled carbon nanotubes (MWCNT). The composites were evaluated for their cure characteristics, mechanical properties, abrasion resistance, compression set, swelling resistance, and crosslinking density. The results indicated that as the hybrid filler content increased, minimum torque, maximum torque, delta torque, and cure rate index decreased, whereas scorch time and optimum cure time increased. Mechanical testing revealed that tensile strength and stress at 100% elongation improved up to a filler concentration of 4 phr but declined beyond this level. Hardness and tear strength showed a continuous increase with higher filler loading, while elongation at break and rebound resilience decreased. Additionally, compression set increased with rising filler content. Swelling resistance and abrasion resistance were optimized at 4 phr of hybrid filler but deteriorated when the filler concentration exceeded this threshold.

Keywords EPDM/SBR · POSS/MWCNT · Swelling resistance · Mechanical properties

1 Introduction

A key challenge in polymer vulcanizates is replacing conventional fillers like carbon black and silica, which are used in large quantities, with smaller amounts of nanofillers such as nanoclay, multiwalled carbon nanotubes (MWCNTs), and graphene to enhance material properties [1–7]. Among the various nanofillers studied, polyhedral oligomeric silsesquioxanes (POSS) and MWCNTs are of particular interest in this research. POSS belongs to a unique class of organic–inorganic hybrid compounds widely applied in nanomaterial and nanocomposite synthesis [8, 9]. Its molecular structure follows the general formula $R_n(SiO_{1.5})_n$, where R represents organic groups, and n varies (typically 8, 10, or 12) depending on its configuration [10]. These cage-like molecules offer exceptional thermal and chemical stability [11]. Although POSS molecules are inherently nanoscale (3–5 nm), they often form micro-sized agglomerates in bulk [12]. However, when dispersed in solvents or polymer matrices, they revert to their nanoscale dimensions. POSS-based nanocomposites have gained attention for their versatile

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